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Technical Design Document (TDD)

Editorial Manager® Peer Review Portal & OJS 3.4 Integration

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System Target: Davao Research Journal (`davaoresearchjournal.ph` / `review.davaoresearchjournal.ph`)
Core Architecture: DRJ Adaptive Reactive Workstation Core (ARWC), TypeScript, Open Journal Systems (OJS 3.4.0-7), PHP 8.2, MySQL 8.0, Resend SMTP Relay.
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1. Executive Summary & Problem Statement

1.1 Background

The standard Open Journal Systems (OJS) 3.4 reviewer workflow requires external reviewers to navigate a multi-step login portal, remember credentials, and submit evaluations in standard form pages with separate tabs for viewing submissions. For busy academic reviewers, this creates significant friction, causing delayed evaluations, unaccepted invitations, and high cognitive load.

1.2 The Solution

The **Editorial Manager® Peer Review Portal** is a decoupled, modern Single Page Application (SPA) styled after commercial enterprise systems (Aries Editorial Manager® and Elsevier EM). It operates on a dedicated subdomain (`review.davaoresearchjournal.ph`), connects directly to the live OJS 3.4 database and file storage, and enforces a zero-trust multi-gate security architecture.

2. High-Level System Architecture

The ecosystem consists of four interconnected tiers: 1. **Frontend Presentation Tier (Client)**: DRJ Adaptive Reactive Workstation Core (ARWC) featuring drag-resizable split panes, reactive scoring rubrics, and three security gates. 2. **Security & Routing Tier**: Nginx web server handling subdomain routing, TLS termination, and frame-0 solid background enforcement. 3. **API & Business Logic Gateway**: Lightweight, high-throughput PHP 8.2 REST API (`api.php`) enforcing HMAC-SHA256 signature validation and payload sanitization. 4. **Data & Storage Tier**: OJS 3.4 core database (`drj2024_35beta`), local file storage repository (`/home/drj2024/files/`), and Resend SMTP Relay (Port 465 SSL).

```
flowchart TD
    subgraph ClientLayer["Frontend Client Tier (Browser)"]
        A["Reviewer Browser / Device"] ⟶|Opens Link| B{"Device Gate Check"}
        B ⟶|Mobile Screen < 1024px| C["⛔ Block Screen: Desktop Required\n(Clipboard Link Co
        B ⟶|Desktop / Tablet ⩾ 1024px| D{"Invitation Token Present?"}
        D ⟶|No Token| E["⛔ Direct Access Restricted Gate\n(Solid #00172B Backdrop)"]
        D ⟶|Token Valid| F{"Email Gate Verified?"}
        F ⟶|No / Cached Mismatch| G["🔒 Identity Verification Challenge\n(Enter Invited Emai
        F ⟶|Verified Match| H["🔓 Aries EM Split-Screen Workspace\n(Manuscript + Evaluation
    end

    subgraph APILayer["API Gateway Tier (review.davaoresearchjournal.ph)"]
        H ⟶|GET api.php?action=get_assignment| API["REST API Gateway (api.php)"]
        H ⟶|POST api.php?action=save_draft| API
        H ⟶|POST api.php?action=submit_review| API
        H ⟶|GET api.php?action=stream_file| API
    end

    subgraph CoreLayer["OJS 3.4 Infrastructure Tier"]
        API ⟶|PDO Connection| DB[("OJS 3.4 Database\n(drj2024_35beta)")]
        API ⟶|Encrypted Path Lookup| FS[("Secure File Storage\n/home/drj2024/files/")]
        OJS["OJS Core Engine & Editorial Board"] ⟶|Review Assignment Created| DB
        OJS ⟶|Review Invitation Dispatch| SMTP["Resend SMTP Relay\n(smtp.resend.com:465 SSL)
        SMTP ⟶|TLS Delivery| EMAIL["Reviewer Gmail / Academic Inbox"]
    end
```

3. Triple-Gate Security Protocol

To ensure confidential, unpublished scientific manuscripts are protected against unauthorized access, URL scraping, and premature phone submissions, three sequential gates must be passed:

```
sequenceDiagram
    autonumber
    actor Reviewer
    participant Client as DRJ Reactive Workstation (ARWC)
    participant API as API Gateway (api.php)
    participant DB as OJS Database
    participant FS as File Storage

    Note over Reviewer,Client: GATE 1: Device Resolution Gate
    Reviewer->>Client: Navigate to review link
    Client->>Client: Inspect navigator.userAgent & window.innerWidth
    alt Screen < 1024px (Mobile Phone)
        Client-->>Reviewer: Display Desktop Required Gate (Block UI)
    else Screen ≥ 1024px (Desktop / Laptop)
        Note over Reviewer,Client: GATE 2: Direct Access Gate
        Client->>Client: Check URL params (reviewId, token)
        alt reviewId or token missing
            Client-->>Reviewer: Display Reviewer Invitation Required Gate (Block UI)
        else reviewId & token present
            Note over Reviewer,FS: GATE 3: Email Verification Challenge
            Client->>API: GET get_assignment (review_id, token, email="")
            API->>DB: Verify HMAC-SHA256 signature for review_id
            API-->>Client: HTTP 200 { requires_email: true, email_hint: "s**t@manusci.com" }
            Client-->>Reviewer: Display Identity Challenge Modal
            Reviewer->>Client: Inputs "support@manusci.com"
            Client->>API: GET get_assignment (review_id, token, email="support@manusci.com")
            API->>DB: Query user email for review assignment
            alt Email matches reviewer record
                API-->>Client: HTTP 200 { success: true, data: assignment_payload }
                Client->>Client: Cache verified email in localStorage
                Client-->>Reviewer: Unlock EM Workspace & Render Manuscript
            else Email does NOT match
                API-->>Client: HTTP 403 { success: false, error: "Access Denied" }
                Client-->>Reviewer: Display Unauthorized Error Alert
            end
        end
    end
```

3.1 Cryptographic Token Specification

- **Algorithm**: HMAC-SHA256
- **Key Derivation**: Server salt derived from OJS `config.inc.php` secret key + application namespace.
- **Payload String**: `{$review_id}|{$date_assigned}`
- **Signature Output**: 64-character lowercase hexadecimal hash.
- **Properties**: Tamper-proof; invalidating either the `review_id` or reassignment timestamp automatically revokes the token.

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4. Split-Screen Workspace & Reactive Scoring Engine

4.1 Component Breakdown

- `App.vue`: Root orchestrator. Controls dynamic layout switching (`split`, `manuscript`, `rubric`) via an Adaptive Dual-Pane Split Controller and enforces frame-0 solid navy backdrop when gates are active.
- `WorkspaceHeader.vue`: Aries EM high-density toolbar. Shows DRJ branding, live MS ID tracking, OJS bridge health pulse, auto-save status indicator, and deadline countdown.
- `ManuscriptViewer.vue`: Left-pane document reader. If PDF proof exists, loads high-performance sandbox iframe streaming from `api.php?action=stream_file`. If Word (.docx), displays a structured manuscript card with title, abstract, and direct file download.
- `ReviewerEvaluationForm.vue`: Right-pane evaluation suite:
- **Scientific Criteria Matrix**: 5-point Likert scale (Scale 1: Poor to 5: Outstanding) across 5 core criteria.
- **Reviewer Questionnaire**: Compliance radios (journal suitability, statistics rigor, language clarity, conflicts of interest).
- **Blinded Comments for Author(s)**: Transmitted directly to authors upon editorial decision.
- **Confidential Comments to Editor-in-Chief**: Strictly concealed from authors; private to editorial board.
- **Reviewer File Attachments**: Drag-and-drop uploader for marked-up manuscript drafts.
- **Final Recommendation**: Accept, Minor Revisions, Major Revisions, or Decline.
- `SubmitConfirmModal.vue`: Pre-submission audit summary preventing accidental submissions.
- `CertificateModal.vue`: Instant generation of an official, printable **Certificate of Peer Review Recognition** with institutional seal.

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5. Real-Time Auto-Save Protocol

To prevent work loss caused by network interruptions or accidental browser closures, the portal uses a debounced auto-save protocol:

```
sequenceDiagram
    autonumber
    actor Reviewer
    participant Store as In-Memory State & Telemetry Store
    participant Local as Browser LocalStorage
    participant API as api.php (save_draft)
    participant FS as Server JSON Draft Storage

    Reviewer-->>Store: Modifies score radio or types in textarea
    Store-->>Store: Update completion percentage reactive meter
    Store-->>Store: Set saveStatus = "Saving draft..."
    Store-->>Store: Debounce timer started (1200ms delay)

    alt Reviewer continues typing within 1200ms
        Reviewer-->>Store: Keypress event
        Store-->>Store: Reset debounce timer
    else Inactivity exceeds 1200ms
        Store-->>API: POST /api.php?action=save_draft { review_id, token, email, draft }
        API-->>API: Validate HMAC token & reviewer identity
        API-->>FS: Write payload to /review_portal/drafts/draft_{review_id}.json
        FS-->>API: Write confirmed
        API-->>Store: HTTP 200 { success: true }
        Store-->>Store: Set saveStatus = "Saved HH:MM:SS"
    end
```

6. Submission & OJS Database Synchronization

Upon final submission confirmation, `api.php` executes an atomic database transaction updating core OJS 3.4 tables:

```
sequenceDiagram
    autonumber
    actor Reviewer
    participant Client as Frontend SPA
    participant API as api.php (submit_review)
    participant DB as MySQL (drj2024_35beta)
    participant Drafts as Server Draft Storage

    Reviewer-->>Client: Clicks "Confirm & Transmit to Editor"
    Client-->>API: POST /api.php?action=submit_review
    API-->>API: Validate token, email, and completion constraints

    API-->>DB: UPDATE review_assignments SET recommendation = ?, date_completed = NOW() WHERE r
    API-->>DB: INSERT INTO submission_comments (author comments, comment_type = 1)
    API-->>DB: INSERT INTO submission_comments (editor confidential comments, comment_type = 2)
    API-->>Drafts: Remove or archive draft_{review_id}.json

    DB-->>API: Transaction committed
    API-->>Client: HTTP 200 { success: true, message: "Review successfully submitted" }
    Client-->>Client: Set assignment.is_completed = true
    Client-->>Reviewer: Display Success Alert & Open Certificate Modal
```

7. OJS 3.4 Generic Plugin Architecture

The OJS generic plugin (`plugins/generic/editorialManagerReview`) enables institutional management and internal routing:

```
plugins/generic/editorialManagerReview/
├── EditorialManagerReviewPlugin.php  # Extends PKP GenericPlugin; registers LoadHandler
├── version.xml                       # OJS 3.4 plugin version metadata
├── index.php                         # PSR-4 plugin bootstrap
├── locale/
│   └── en/
│       └── locale.po                 # English localization strings
└── pages/
    └── ReviewerPortalHandler.php     # Intercepts /index.php/drj/reviewer-portal
```

7.1 Lifecycle Hook Integration

```
public function register($category, $path, $mainContextId = null): bool
{
    $success = parent::register($category, $path, $mainContextId);
    if ($success) {
        Hook::add('LoadHandler', $this→callbackLoadHandler(...), Hook::SEQUENCE_CORE);
    }
    return $success;
}
```

When OJS receives an inbound request matching `reviewer-portal` or `em-review`, the plugin delegates handling to `ReviewerPortalHandler`, which transparently preserves all query parameters (`reviewId`, `token`) and transitions the reviewer to the dedicated portal environment.

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8. Network Infrastructure & SMTP Relay

- **Hosting Server**: GoDaddy VPS Gen 4 (AlmaLinux 9.8, cPanel user `drj2024`).
- **Port 25 Policy**: Outbound TCP port 25 is blocked by network-level firewall. Direct sendmail/Exim connections to Google/Yahoo time out.
- **Resend SMTP Relay**: Outbound review invitations route through `smtp.resend.com` on **Port 465 SSL**:
- Host: `smtp.resend.com`
- Port: `465` (SSL)
- Default Envelope Sender: `editor@davaoresearchjournal.ph`
- Deliverability: SPF/DKIM authenticated; instantaneous delivery to external Gmail inboxes.

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9. Verification & Audit Trail

Test Case	Scenario	Expected Outcome	Verified
TC-01	Visit `review.davaoresearchjournal.ph` without parameters	Fullscreen solid navy backdrop; Reviewer Invitation Required gate displayed; zero template leakage.	**PASSED**
TC-02	Open review link on smartphone (< 1024px width)	Device Gate active; displays desktop requirement notice and 1-click clipboard copy button.	**PASSED**
TC-03	Open review link on desktop with valid token	Identity Verification Gate displayed; prompts for assigned email; manuscript concealed.	**PASSED**
TC-04	Input wrong reviewer email	Access denied alert; manuscript remains cryptographically locked.	**PASSED**
TC-05	Input correct reviewer email (`support@manusci.com`)	Gate unlocks immediately; manuscript PDF / Word rendered; auto-save engine activates.	**PASSED**
TC-06	Review submission transmittal	OJS `review_assignments` updated (`date_completed = NOW()`); comments recorded; Certificate modal opens.	**PASSED**
TC-07	Outbound Invitation Email	Dispatched through Resend SMTP port 465 SSL to external Gmail inbox.	**PASSED**